Abbilash Shivaparan

Princeton, NJ · abbi.shiva@gmail.com · linkedin.com/in/abbi-shiva · github.com/abbishiva-oss

Information Science student at the University of Maryland pursuing a minor in Data Science, with analytics experience spanning SQL, Excel reporting, Power BI dashboarding, and Python-based automation for data cleaning and processing. Developed full-stack web applications and machine learning projects, and gained applied experience with retrieval-augmented generation (RAG) systems, large language models, and natural language processing through an industry AI externship and undergraduate research.

EXPERIENCE

Pfizer (via Extern) · *AI-Powered Document Insights Extern* 06/2026 – 08/2026

- Selected for 10-week externship building AI document-analysis skills across RAG systems, large language models, NLP, and prompt engineering.
- Built a RAG pipeline over pharmaceutical certificate documents; resolved corrupted PDF extraction by switching to PyMuPDF and tuned retrieval depth to recover multi-section answers.

NJ TRANSIT · *Data Analytics – IT Intern* 05/2024 – 09/2025

- Built SQL queries and Excel reports generating weekly operational statistics for 5,000+ ride records; automated geospatial validation of 2,000+ addresses using Haversine calculations, reducing manual review by 60%.
- Developed Python scripts using Google Reverse Geocoding API to rank top 25 high-frequency pickup locations for route optimization.
- Built 3 Power BI dashboards visualizing ridership trends and operational KPIs; created Crystal Reports dashboard flagging 150+ mismatched location IDs and presented enhancement roadmap to management.

Windsor Internal Medicine and Geriatrics · *Office Assistant – IT Support* 06/2022 – Present

- Implemented access controls and security protocols for patient data protection; performed Windows patching across 10+ workstations maintaining HIPAA compliance.
- Resolved OS compatibility issues on corporate PCs, minimizing clinical staff downtime.

RESEARCH

University of Maryland, Chan Lab · *Undergraduate Researcher (INST408G), Mentor: Joel Chan* 09/2026 – Present

- Evaluating how accurately frontier and local LLMs extract claims and evidence from research papers against human review, using 205 model-drafted discourse graph nodes across 27 papers.
- Leading local model setup and scoring (precision, recall, F1) and categorizing model errors by type.

PROJECTS

DataProcessBot · RAG desktop app for document Q&A (Tauri, Python FastAPI, ChromaDB)

- Built a cross-platform desktop app that lets users query their own documents through a retrieval-augmented generation pipeline, with auto-summarize, extract, and flag actions on upload.

SubleaseHer · Full-stack sublease marketplace (Next.js, PostgreSQL, Prisma, NextAuth.js, Tailwind CSS) subleaseher.vercel.app

- Designed and deployed marketplace connecting students with housing; engineered PostgreSQL schema with Prisma ORM, integrated Cloudinary for images, and implemented NextAuth.js authentication.

NBA Player DNA · ML clustering dashboard (Python, scikit-learn, Streamlit) nba-player-dna.streamlit.app

- Segmented 500+ players into 8 archetypes using PCA and K-Means; built interactive Streamlit dashboard, updated with 2025-26 season data.

Electric Vehicle Registration Analysis · SQL, 3NF normalization, ER diagrams

- Designed normalized database for 100K+ WA State EV records; found top 3 manufacturers account for 78% of registrations.

EDUCATION

University of Maryland, College Park · B.S. Information Science, Minor in Data Science Expected May 2027

Activities: Cyber Security Club, AWS Practitioner Club, UMD Club Basketball

SKILLS & CERTIFICATIONS

Data & BI: SQL, Excel (PivotTables, XLOOKUP), Power BI, Crystal Reports

Programming: Python, R, Java, JavaScript, TypeScript, React, Next.js

AI & ML: RAG, LLMs, LlamaIndex, ChromaDB, scikit-learn, pandas, PCA, K-Means, Streamlit

Web & Tools: PostgreSQL, Prisma, FastAPI, Tauri, Tailwind CSS, Git, GitHub, Figma

Certificates: Career Essentials in Data Analysis (Microsoft), Data Analytics Specialist (LinkedIn Learning)